In [1]:
```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

In [2]:
```
df = pd.read_csv("WA_Fn-UseC_-HR-Employee-Attrition.csv")
df.head()
```

Out[2]:

	Age	Attrition	BusinessTravel	DailyRate	Department	DistanceFromHome	Educat
0	41	Yes	Travel_Rarely	1102	Sales	1	
1	49	No	Travel_Frequently	279	Research & Development	8	
2	37	Yes	Travel_Rarely	1373	Research & Development	2	
3	33	No	Travel_Frequently	1392	Research & Development	3	
4	27	No	Travel_Rarely	591	Research & Development	2	

5 rows × 35 columns

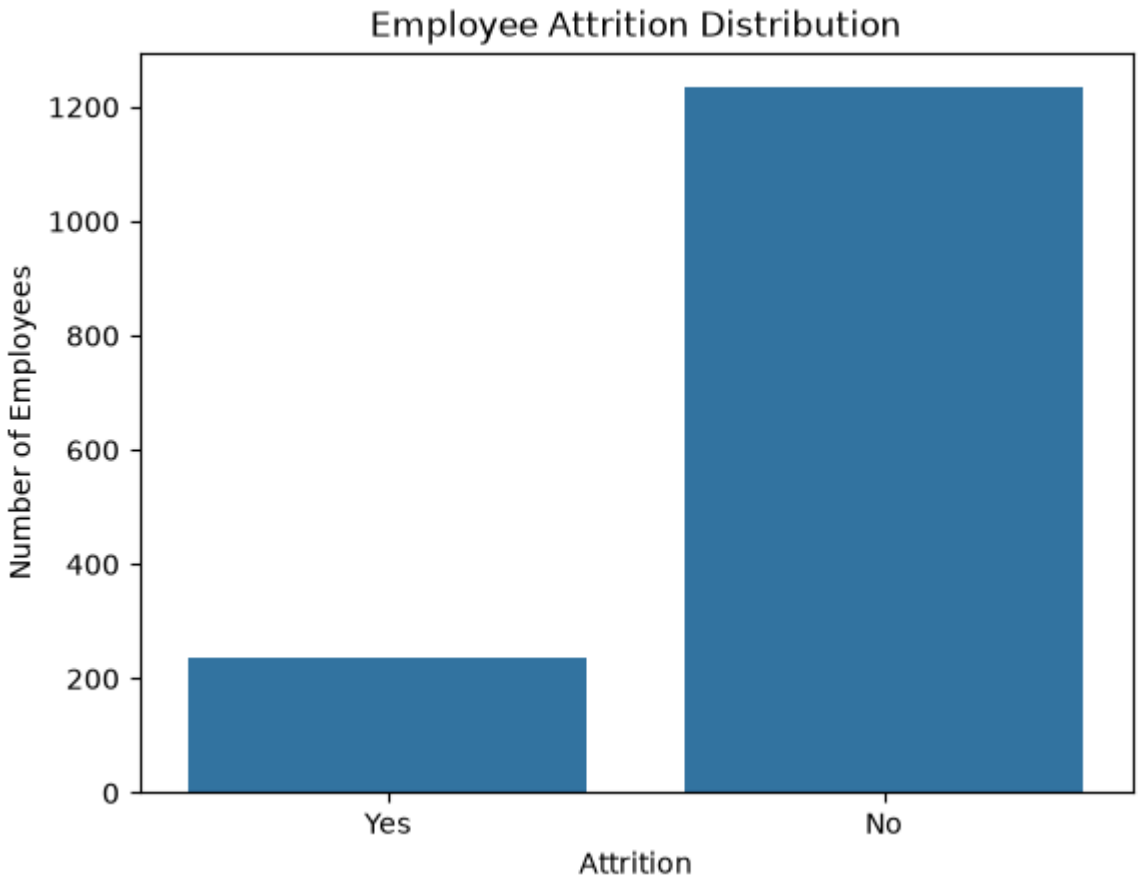
In [3]:
```
# Count employees who stayed and left
print(df["Attrition"].value_counts())

# Calculate total employees and attrition rate
total_employees = len(df)
employees_left = (df["Attrition"] == "Yes").sum()
active_employees = (df["Attrition"] == "No").sum()
attrition_rate = (employees_left / total_employees) * 100

print("Total Employees:", total_employees)
print("Employees Left:", employees_left)
print("Active Employees:", active_employees)
print("Attrition Rate:", round(attrition_rate, 2), "%")
```

```
Attrition
No     1233
Yes     237
Name: count, dtype: int64
Total Employees: 1470
Employees Left: 237
Active Employees: 1233
Attrition Rate: 16.12 %
```

In [4]:
```
sns.countplot(data=df, x="Attrition")
plt.title("Employee Attrition Distribution")
plt.xlabel("Attrition")
plt.ylabel("Number of Employees")
plt.show()
```

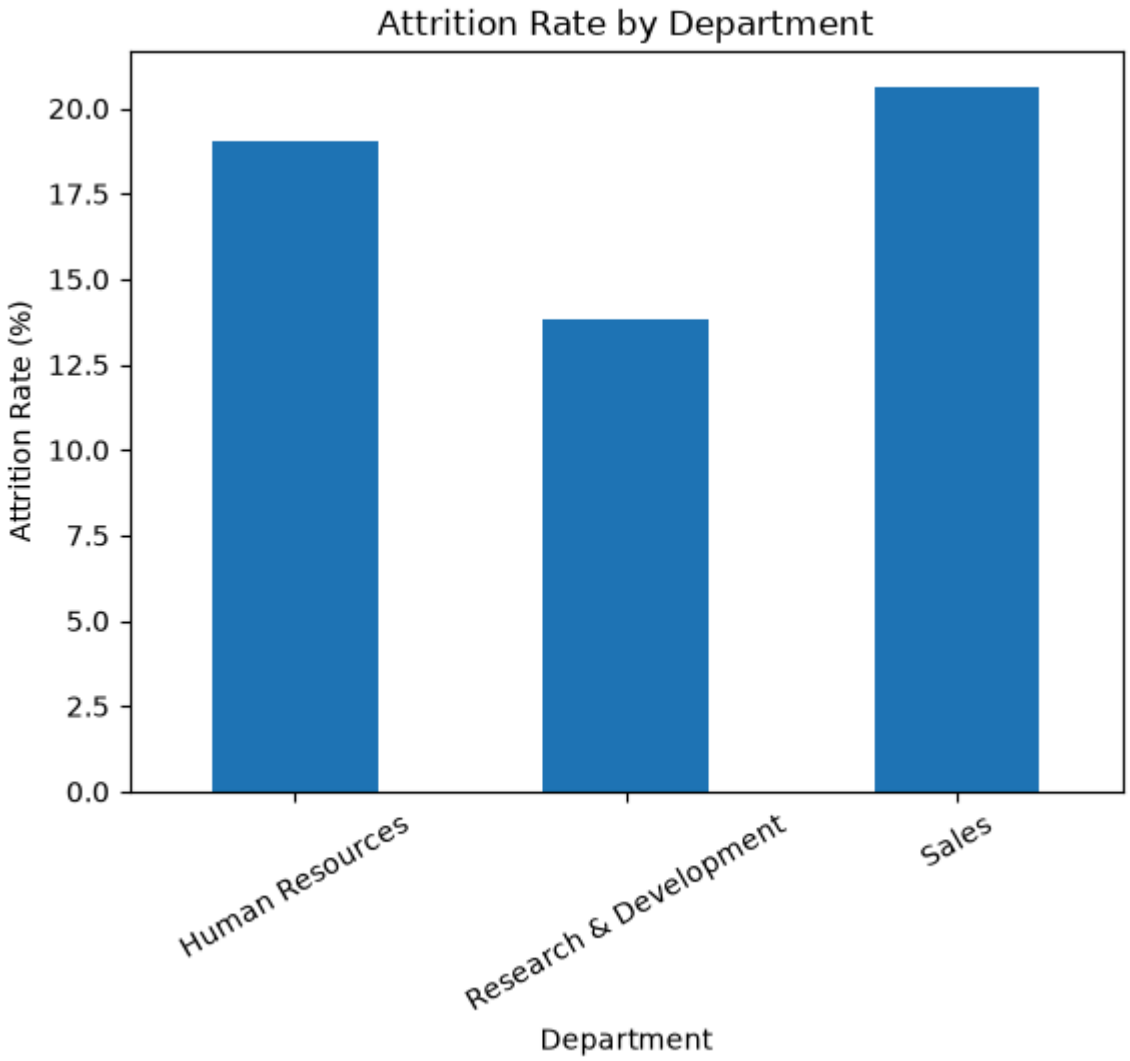


In [5]:
```
department_attrition = pd.crosstab(
    df["Department"], df["Attrition"], normalize="index"
) * 100

display(department_attrition)

department_attrition["Yes"].plot(kind="bar")
plt.title("Attrition Rate by Department")
plt.xlabel("Department")
plt.ylabel("Attrition Rate (%)")
plt.xticks(rotation=30)
plt.show()
```

Attrition	No	Yes
Department		
Human Resources	80.952381	19.047619
Research & Development	86.160250	13.839750
Sales	79.372197	20.627803

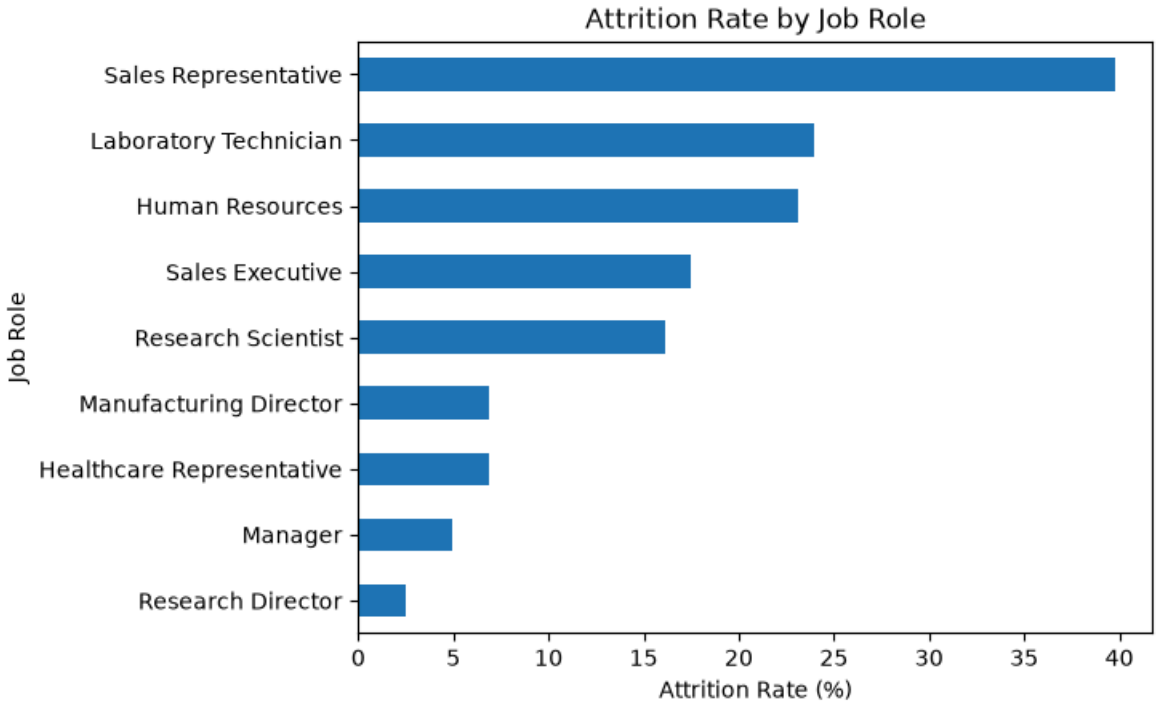


In [6]:
```
jobrole_attrition = pd.crosstab(
    df["JobRole"], df["Attrition"], normalize="index"
) * 100

display(jobrole_attrition)

jobrole_attrition["Yes"].sort_values().plot(kind="barh")
plt.title("Attrition Rate by Job Role")
plt.xlabel("Attrition Rate (%)")
plt.ylabel("Job Role")
plt.show()
```

Attrition	No	Yes
JobRole		
Healthcare Representative	93.129771	6.870229
Human Resources	76.923077	23.076923
Laboratory Technician	76.061776	23.938224
Manager	95.098039	4.901961
Manufacturing Director	93.103448	6.896552
Research Director	97.500000	2.500000
Research Scientist	83.904110	16.095890
Sales Executive	82.515337	17.484663
Sales Representative	60.240964	39.759036



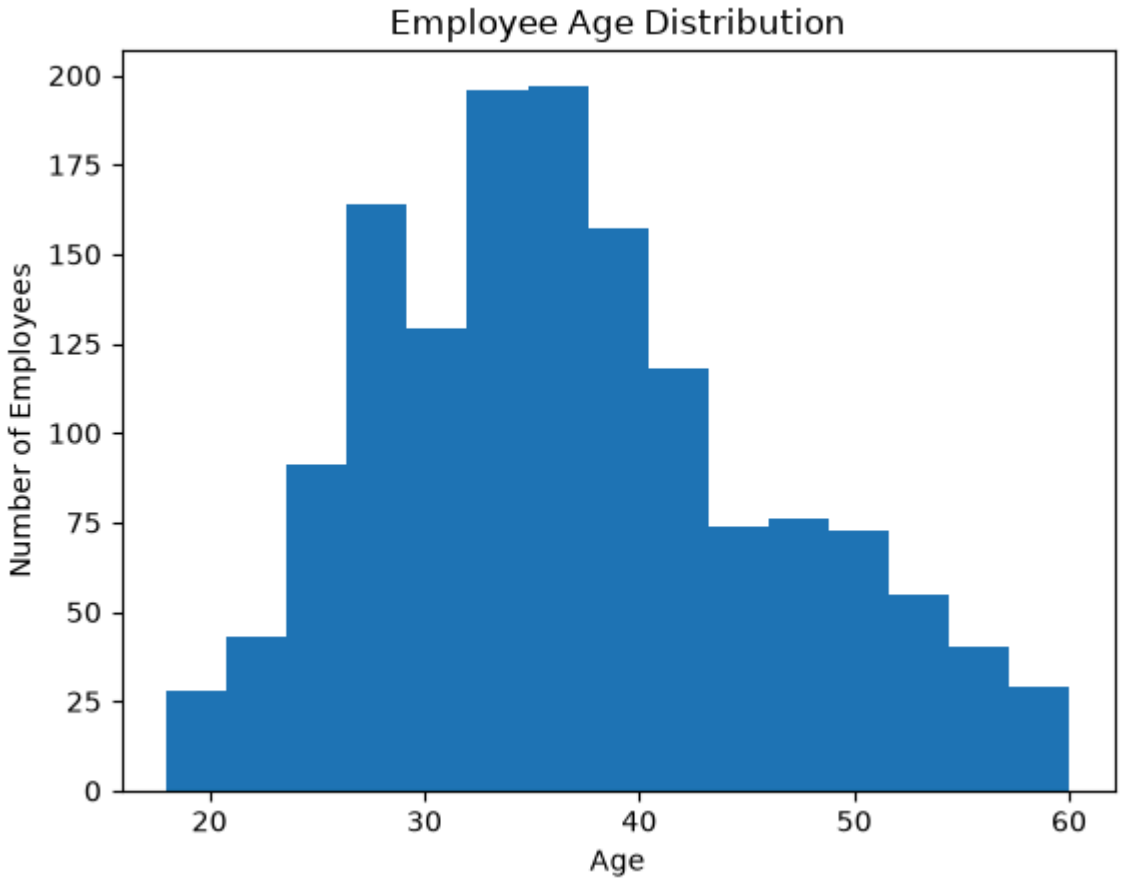
In [7]:
```
df["Age"].plot(kind="hist", bins=15)
plt.title("Employee Age Distribution")
plt.xlabel("Age")
plt.ylabel("Number of Employees")
plt.show()

df["AgeGroup"] = pd.cut(
    df["Age"],
    bins=[17, 25, 35, 45, 55, 65],
    labels=["18-25", "26-35", "36-45", "46-55", "56-65"]
)

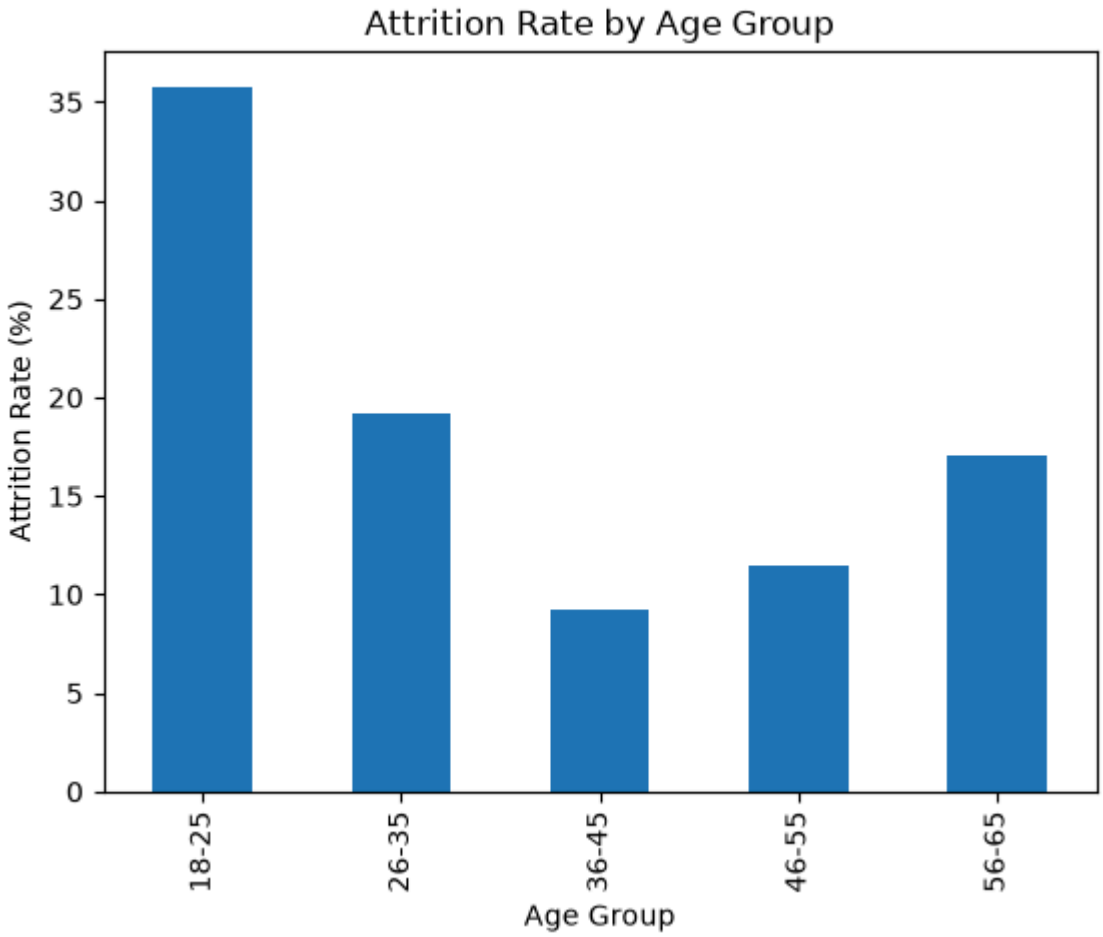
age_attrition = pd.crosstab(
    df["AgeGroup"], df["Attrition"], normalize="index"
) * 100

display(age_attrition)
```

```
age_attrition["Yes"].plot(kind="bar")
plt.title("Attrition Rate by Age Group")
plt.xlabel("Age Group")
plt.ylabel("Attrition Rate (%)")
plt.show()
```



Attrition	No	Yes
AgeGroup		
18-25	64.227642	35.772358
26-35	80.858086	19.141914
36-45	90.811966	9.188034
46-55	88.495575	11.504425
56-65	82.978723	17.021277

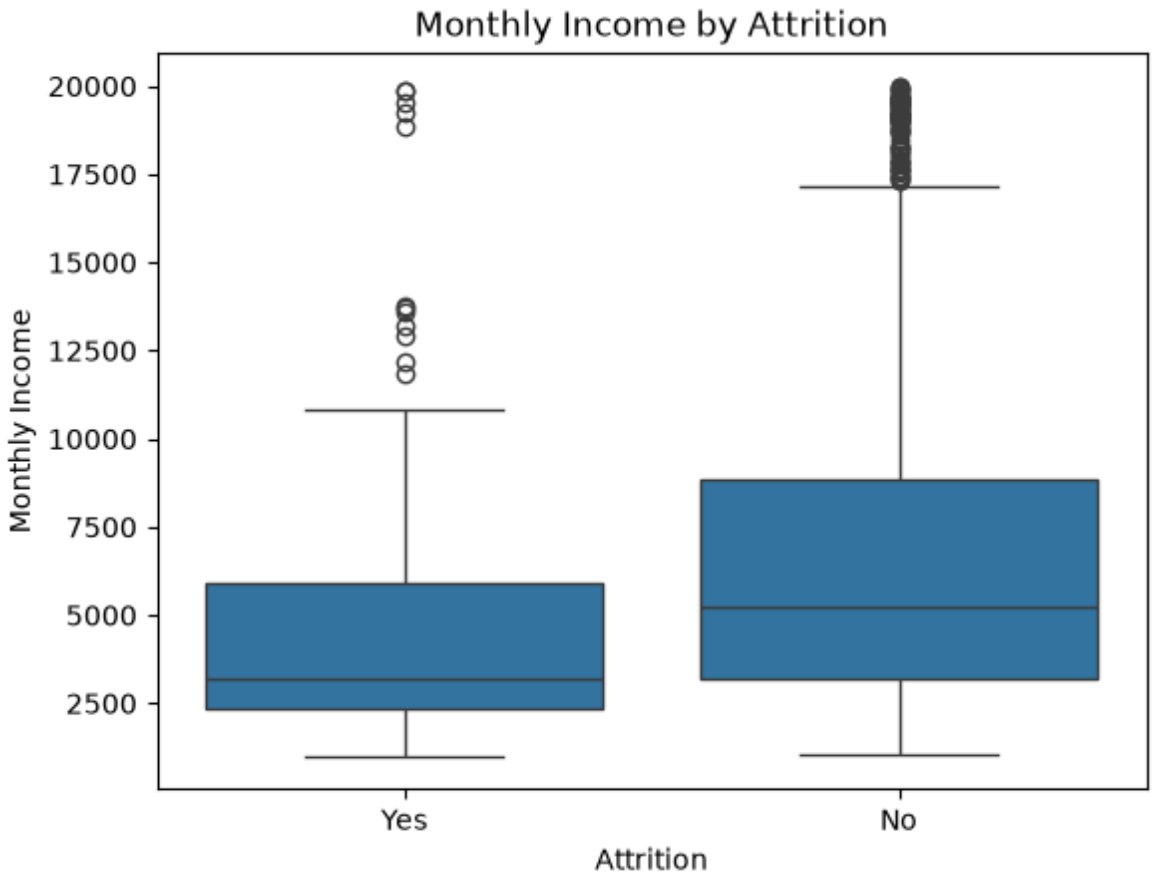


In [8]:
```
salary_summary = df.groupby("Attrition")["MonthlyIncome"].mean()

print("Average Monthly Income by Attrition:")
display(salary_summary)

sns.boxplot(data=df, x="Attrition", y="MonthlyIncome")
plt.title("Monthly Income by Attrition")
plt.xlabel("Attrition")
plt.ylabel("Monthly Income")
plt.show()
```

```
Average Monthly Income by Attrition:
Attrition
No     6832.739659
Yes    4787.092827
Name: MonthlyIncome, dtype: float64
```

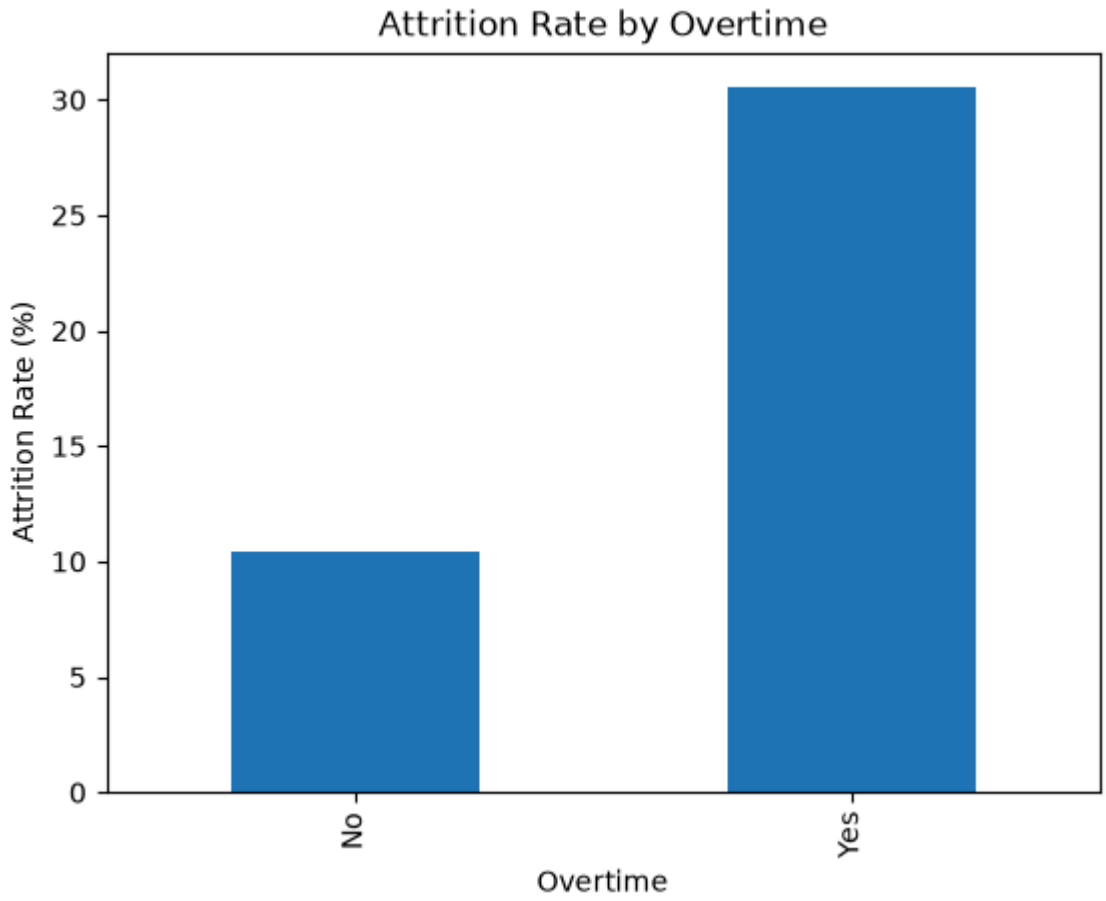


In [9]:
```
overtime_attrition = pd.crosstab(
    df["OverTime"], df["Attrition"], normalize="index"
) * 100

display(overtime_attrition)

overtime_attrition["Yes"].plot(kind="bar")
plt.title("Attrition Rate by Overtime")
plt.xlabel("Overtime")
plt.ylabel("Attrition Rate (%)")
plt.show()
```

Attrition	No	Yes
OverTime		
No	89.563567	10.436433
Yes	69.471154	30.528846

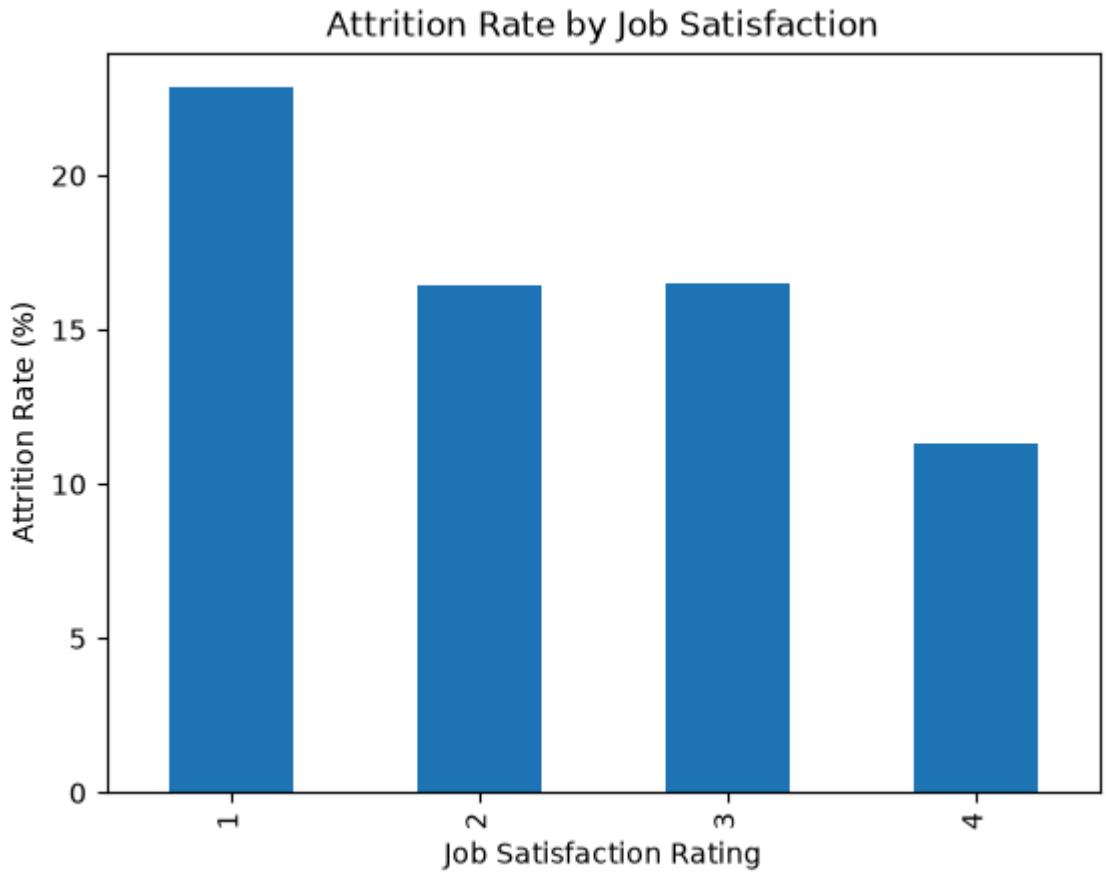


In [10]:
```
satisfaction_attrition = pd.crosstab(
    df["JobSatisfaction"], df["Attrition"], normalize="index"
) * 100

display(satisfaction_attrition)

satisfaction_attrition["Yes"].plot(kind="bar")
plt.title("Attrition Rate by Job Satisfaction")
plt.xlabel("Job Satisfaction Rating")
plt.ylabel("Attrition Rate (%)")
plt.show()
```

Attrition	No	Yes
JobSatisfaction		
1	77.162630	22.837370
2	83.571429	16.428571
3	83.484163	16.515837
4	88.671024	11.328976

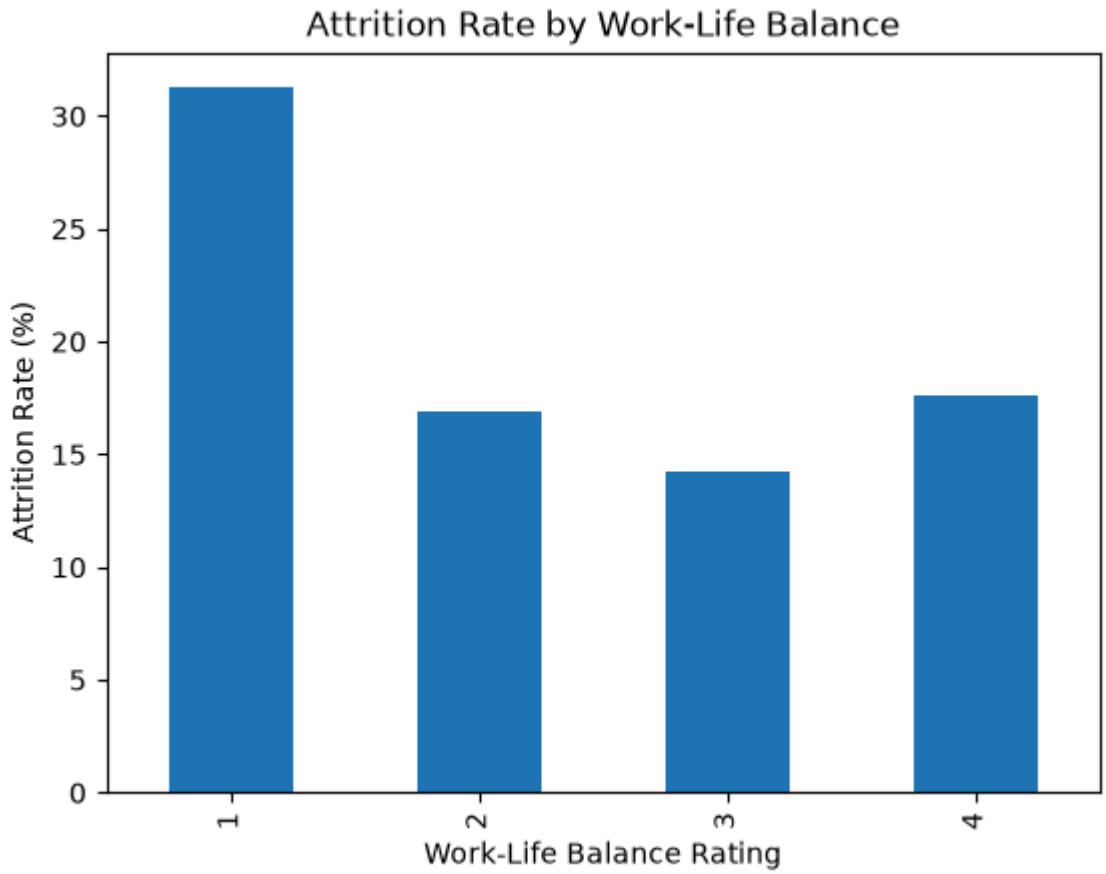


In [11]:
```
worklife_attrition = pd.crosstab(
    df["WorkLifeBalance"], df["Attrition"], normalize="index"
) * 100

display(worklife_attrition)

worklife_attrition["Yes"].plot(kind="bar")
plt.title("Attrition Rate by Work-Life Balance")
plt.xlabel("Work-Life Balance Rating")
plt.ylabel("Attrition Rate (%)")
plt.show()
```

Attrition	No	Yes
WorkLifeBalance		
1	68.750000	31.250000
2	83.139535	16.860465
3	85.778275	14.221725
4	82.352941	17.647059



In [12]:
```
gender_attrition = pd.crosstab(
    df["Gender"], df["Attrition"], normalize="index"
) * 100

marital_attrition = pd.crosstab(
    df["MaritalStatus"], df["Attrition"], normalize="index"
) * 100

print("Attrition by Gender (%)")
display(gender_attrition)

print("Attrition by Marital Status (%)")
display(marital_attrition)
```

```
Attrition by Gender (%)
```

Attrition	No	Yes
Gender		
Female	85.204082	14.795918
Male	82.993197	17.006803

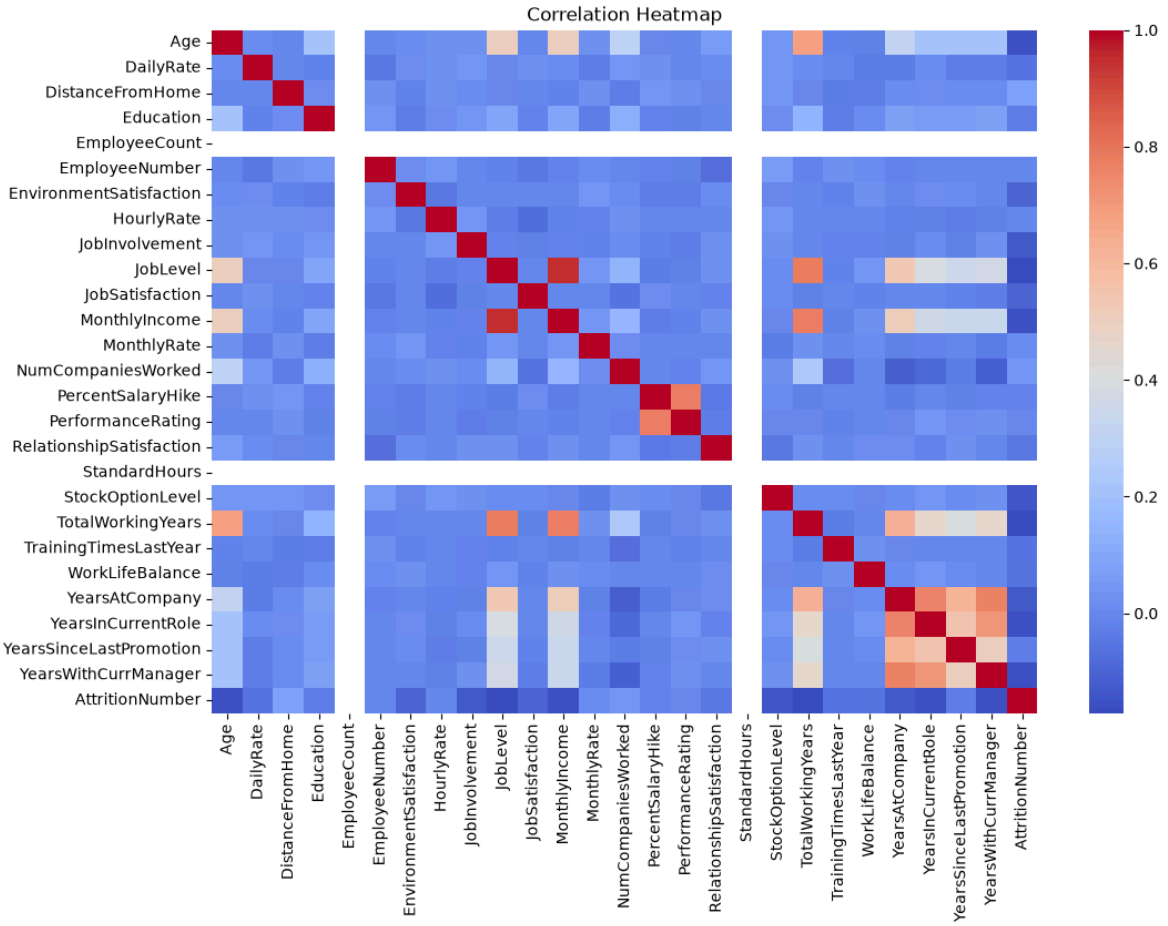
```
Attrition by Marital Status (%)
```

Attrition	No	Yes
MaritalStatus		
Divorced	89.908257	10.091743
Married	87.518574	12.481426
Single	74.468085	25.531915

In [13]:
```
# Convert Attrition into numbers for correlation
df["AttritionNumber"] = df["Attrition"].map({"No": 0, "Yes": 1})

correlation = df.select_dtypes(include="number").corr()

plt.figure(figsize=(12, 8))
sns.heatmap(correlation, cmap="coolwarm")
plt.title("Correlation Heatmap")
plt.show()
```



In [14]:
```
# Save important tables for the final project report
department_attrition.to_csv("attrition_by_department.csv")
jobrole_attrition.to_csv("attrition_by_jobrole.csv")
age_attrition.to_csv("attrition_by_age_group.csv")
overtime_attrition.to_csv("attrition_by_overtime.csv")
satisfaction_attrition.to_csv("attrition_by_job_satisfaction.csv")
worklife_attrition.to_csv("attrition_by_worklife_balance.csv")

# Create a simple KPI report
kpi_report = pd.DataFrame({
    "Metric": [
```

```
        "Total Employees",
        "Employees Left",
        "Active Employees",
        "Attrition Rate (%)",
        "Average Monthly Income"
    ],
    "Value": [
        total_employees,
        employees_left,
        active_employees,
        round(attrition_rate, 2),
        round(df["MonthlyIncome"].mean(), 2)
    ]
})

display(kpi_report)
kpi_report.to_csv("eda_kpi_report.csv", index=False)

print("EDA tables and KPI report saved successfully.")
```

	Metric	Value
0	Total Employees	1470.00
1	Employees Left	237.00
2	Active Employees	1233.00
3	Attrition Rate (%)	16.12
4	Average Monthly Income	6502.93

```
EDA tables and KPI report saved successfully.
```